

PMDD LINGUISTIC ANALYSIS REPORT

Pragmatic Meaning Drift Detector — 5-Agent AI Analysis System

CORPUS: [Corpus Name / Source]
LANGUAGE: English (L1/L2 specify)
SEGMENTS ANALYZED: [N]
TARGET KEYWORDS: [k_1 , k_2 , ...]

ANALYSIS DATE: [YYYY-MM-DD]
REPORT ID: PMDD-[XXXXXX]
AGENTS DEPLOYED: 5 (GPT-4o backbone)
MEMORY MODE: Episodic + Semantic

A1

Corpus Preprocessor

A2

Pragmatic Analyzer

A3

Semantic & Register

A4

Corpus Statistician

A5

Orchestrator Reporter

74OVERALL MEANING DRIFT
(0–100)**81**

PRAGMATIC DRIFT SCORE

68

SEMANTIC FIELD DRIFT

73

REGISTER DRIFT SCORE

Report Purpose: This document constitutes a formal linguistic research output generated by the PMDD 5-Agent AI system. All findings are grounded in established linguistic theories (Grice 1975; Austin 1962; Searle 1969; Lyons 1977; Halliday 1978; Sinclair 1991; Fairclough 1992) and are supported by direct corpus evidence. The report is designed for use by the academic research community.

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SECTION 1 — EXECUTIVE SUMMARY & CORPUS OVERVIEW

This report presents the findings of a full multi-layer linguistic analysis performed by the Pragmatic Meaning Drift Detector (PMDD) system on the submitted text corpus. The analysis is conducted across five interdependent analytical layers — pragmatics, semantic fields, register, corpus statistics, and critical discourse synthesis — each operationalised through a dedicated AI agent with an OpenAI GPT-4o reasoning backbone and episodic memory access. Findings are reported with direct corpus evidence at a minimum 60% evidence-to-interpretation ratio throughout this document.

1.1 Corpus Statistics Dashboard

Metric	Value	Agent Responsible	Linguistic Significance
Total Corpus Segments	[N]	Agent 1	Unit of analysis for all downstream agents
Total Word Tokens	[N]	Agent 1	Corpus size determines statistical power
Type-Token Ratio (TTR)	[0.00–1.00]	Agent 4	Lexical richness indicator (Sinclair 1991)
Segments: Period 1	[N]	Agent 1	Temporal sub-corpus for comparative drift analysis
Segments: Period 2	[N]	Agent 1	Temporal sub-corpus for comparative drift analysis
Speech Acts Labeled	[N] ([X]%)	Agent 2	Coverage rate ≥80% required for validity
Semantic Fields Detected	[N]	Agent 3	Breadth of semantic domain coverage
Collocations Computed	[N] pairs	Agent 4	Lexical co-occurrence network density
Keyness Scores (p<0.05)	[N] items	Agent 4	Statistically significant lexical markers

Table 1.1: Corpus statistics generated by Agent 1 (Corpus Preprocessor) and Agent 4 (Corpus Statistician). All values are drawn from the submitted corpus and verified through the Agent 5 orchestrator quality gate.

1.2 Meaning Drift Score Summary

The Meaning Drift Score is a composite index (0–100) computed by Agent 5 across three sub-dimensions. A score above 60 indicates substantial drift warranting further academic investigation. Sub-scores are derived from statistical thresholds and GPT-4o reasoning chains grounded in the theoretical frameworks listed in Section 5.

Dimension	Score	Interpretation	Primary Theory	Agent
Pragmatic Drift	[XX]/100	Shift in speech act ratios & maxim violation frequency	Grice (1975); Searle (1969)	A2
Semantic Field Drift	[XX]/100	Vocabulary movement across semantic domains	Lyons (1977); CDA Fairclough	A3
Register Drift	[XX]/100	Formal/informal register borrowing detected	Halliday (1978) — Tenor/Field/Mode	A3
COMPOSITE SCORE	[XX]/100	Weighted average; threshold 60 = significant	All frameworks	A5

SECTION 2 — PRAGMATIC ANALYSIS (Agent 2 Findings)

Theoretical Grounding: Speech Act Theory (Austin 1962; Searle 1969), Grice's Cooperative Principle & Maxims (1975), Politeness Theory (Brown & Levinson 1987). Agent 2 applies these frameworks adaptively using GPT-4o reasoning — not hard-coded rules — enabling identification of indirect and embedded speech acts that NLP rule-based systems fail to capture.

2.1 Speech Act Distribution with Direct Corpus Evidence

The following table presents the distribution of Searle's five illocutionary speech act categories across the two corpus periods. Percentage change columns identify statistically significant shifts in discourse function. A minimum of three corpus examples per category are cited as direct evidence.

Speech Act (Searle)	Period 1 (%)	Period 2 (%)	Δ Change	Significance	Example Seg IDs
Assertive	[XX]%	[XX]%	[±XX]%	p < [0.05]	#[N], #[N], #[N]
Directive	[XX]%	[XX]%	[±XX]%	p < [0.05]	#[N], #[N], #[N]
Commissive	[XX]%	[XX]%	[±XX]%	p < [0.05]	#[N], #[N], #[N]
Expressive	[XX]%	[XX]%	[±XX]%	p < [0.05]	#[N], #[N], #[N]
Declaration	[XX]%	[XX]%	[±XX]%	p < [0.05]	#[N], #[N], #[N]

Table 2.1: Speech act distribution computed over [N] segments. Significance tested using chi-square (χ^2) against expected equal distribution. Segment IDs link directly to raw corpus evidence in Appendix B.

Corpus Evidence — Speech Act Shift (Illustrative Structure)

Each finding below follows the required evidence structure: **(1) raw corpus segment** → **(2) agent annotation** → **(3) theoretical interpretation**. This 3-layer structure ensures the research community can trace every analytical claim back to its primary corpus source.

SEG #203 — CORPUS EVIDENCE

"We must unite our communities against this threat and take decisive action before the window of opportunity closes permanently."

Agent 2 Annotation → Speech Act: **DIRECTIVE** (Commissive sub-type) | Maxim Violation: **Quantity (over-informative)** | Politeness Score: **2/5** (face-threatening — negative politeness) | Implicature: **Conventional** — urgency encoded via temporal boundary phrase.

Theoretical Interpretation (Agent 5 Synthesis): The co-occurrence of a directive illocution with a Quantity violation and negative politeness strategy is consistent with Grice's (1975) observation that cooperative maxim flouting generates additional implicatures. In this instance, the over-informative structure amplifies urgency — a finding replicated across [N] segments in Period 2 (cf. Segments #445, #612).

SEG #445 — CORPUS EVIDENCE

"The data clearly shows — as every responsible stakeholder already knows — that our policy framework requires immediate revision."

Agent 2 Annotation → Speech Act: **ASSERTIVE** (with embedded Directive) | Maxim Violation: **Relation (implicature mismatch)** | Politeness Score: **3/5** | Implicature: **Conversational** — presupposes shared knowledge to bypass Quality maxim.

Theoretical Interpretation (Agent 5 Synthesis): The embedding of a directive within an assertive surface structure constitutes an indirect speech act (Searle 1969, pp.59–60). The appeal to shared epistemic ground ('as every responsible stakeholder already knows') is a face-saving strategy (Brown & Levinson 1987) that simultaneously presupposes agreement, suppressing potential face-threat while maintaining directive force.

2.2 Gricean Maxim Violation Analysis

Maxim violation frequency is a key indicator of pragmatic drift. The table below presents violation counts per maxim per corpus period, with chi-square significance values. Violation rates above 15% per period indicate systemic discourse strategy shifts rather than incidental usage.

Maxim (Grice 1975)	Period 1 (n)	Period 2 (n)	Rate Δ	χ^2 Value	p-value
Quantity (over/under-informative)	[N]	[N]	[+/-X]%	[X.XX]	[0.0XX]
Quality (unverifiable/false claim)	[N]	[N]	[+/-X]%	[X.XX]	[0.0XX]
Relation (irrelevance / topic drift)	[N]	[N]	[+/-X]%	[X.XX]	[0.0XX]
Manner (ambiguity / obscurity)	[N]	[N]	[+/-X]%	[X.XX]	[0.0XX]
TOTAL VIOLATIONS	[N]	[N]	[+/-X]%	—	—

SECTION 3 — SEMANTIC FIELD & REGISTER ANALYSIS (Agent 3)

Theoretical Grounding: Semantic Field Theory (Lyons 1977), Register Analysis — Tenor, Field, Mode (Halliday 1978 SFL), Critical Discourse Analysis (Fairclough 1992). Agent 3 uses GPT-4o semantic clustering — adaptive, not dictionary-based — allowing emergence of novel semantic domains not anticipated at design time.

3.1 Semantic Drift Map — Vocabulary Migration Across Fields

The Semantic Drift Map tracks how target lexical items shift their primary semantic field membership across corpus sections. A word appearing in Field A in Period 1 and Field B in Period 2 constitutes evidence of semantic drift (Lyons 1977, p.250). The table below presents the top drifting words with their field transitions and corpus frequency.

Target Word	Field: Period 1	Field: Period 2	Freq P1	Freq P2	Drift Type	Evidence Segments
community	SOLIDARITY	EXCLUSION	[N]	[N]	Connotative reversal	#203, #445, #612
power	AUTHORITY	CONFLICT	[N]	[N]	Domain expansion	#[N], #[N]
change	PROGRESS	THREAT	[N]	[N]	Evaluative inversion	#[N], #[N]
[keyword]	[Field]	[Field]	[N]	[N]	[Type]	#[N]
[keyword]	[Field]	[Field]	[N]	[N]	[Type]	#[N]

Table 3.1: Semantic drift map for top-[N] drifting words. Drift Type classification follows Lyons' taxonomy of semantic change (amelioration, pejoration, narrowing, broadening, semantic shift).

Corpus Evidence — Semantic Field Migration

SEG #612 — CORPUS EVIDENCE

"Our community has always protected its values from those who wish to dilute them."

Agent 3 Annotation → Target word: 'community' | Semantic Field (Period 2): **EXCLUSION / BOUNDARY-MARKING** | Co-text markers: 'protected', 'values', 'dilute' — defensive, territorial collocates | Register: **Formal-Political** | Register Borrowing: None detected.

Theoretical Interpretation (Agent 5): The clustering of 'community' with territorial and defensive co-text (confirmed by MI scores in Section 4.2) constitutes semantic drift from the SOLIDARITY field (Segment #101, Period 1) to EXCLUSION (Lyons 1977). This migration carries ideological implications consistent with Fairclough's (1992) observation that vocabulary choice naturalises social identities and power relations in discourse.

3.2 Register Analysis — Halliday's Tenor, Field, Mode

Register classification is applied per segment across three Hallidayan dimensions: Tenor (role relationships), Field (domain of activity), and Mode (channel/medium). Register borrowing — defined as formal vocabulary appearing in informal register or vice versa — is flagged as a discourse strategy marker.

Register Category	Period 1 (%)	Period 2 (%)	Shift Δ	Tenor Marker	Corpus Evidence
Formal-Academic	[XX]%	[XX]%	[±X]%	Third person, passive	#[N], #[N]

Register Category	Period 1 (%)	Period 2 (%)	Shift Δ	Tenor Marker	Corpus Evidence
Formal-Political	[XX]%	[XX]%	[±X]%	We-inclusive, imperative	#[N], #[N]
Semi-Formal	[XX]%	[XX]%	[±X]%	Second person address	#[N], #[N]
Informal-Colloquial	[XX]%	[XX]%	[±X]%	Contractions, hedges	#[N], #[N]
Technical	[XX]%	[XX]%	[±X]%	Domain jargon density	#[N], #[N]
Register Borrowing (flagged)	[N] cases	[N] cases	[±N]	Cross-register lexis	#[N], #[N]

SECTION 4 — CORPUS STATISTICS (Agent 4 Findings)

Theoretical Grounding: Corpus Linguistics methodology (Sinclair 1991), Keyness analysis (Scott 1997), Mutual Information scoring (Church & Hanks 1990), Type-Token Ratio for lexical richness, N-gram extraction for formulaic language. All statistics are computed over raw tokenised corpus — not GPT-inferred values.

4.1 Frequency & Keyness Tables

Keyness measures which words are statistically more frequent in one sub-corpus relative to a reference corpus (Scott 1997). A positive keyness value indicates over-use in Period 2; negative indicates under-use. Only items with $p < 0.05$ (log-likelihood test) are reported.

Rank	Keyword	Freq P1	Freq P2	Keyness Score	Direction	Semantic Field (A3)
1	[word]	[N]	[N]	[+XX.X]	Over-used P2 ↑	[Field]
2	[word]	[N]	[N]	[+XX.X]	Over-used P2 ↑	[Field]
3	[word]	[N]	[N]	[-XX.X]	Under-used P2 ↓	[Field]
4	[word]	[N]	[N]	[+XX.X]	Over-used P2 ↑	[Field]
5	[word]	[N]	[N]	[-XX.X]	Under-used P2 ↓	[Field]
...	...	...	...	...	...	Top [N] items total

Table 4.1: Top-[N] keyness items (log-likelihood, $p < 0.05$). Semantic Field column links statistical evidence directly to Agent 3 findings, enabling cross-agent convergence testing.

4.2 Collocation & Mutual Information (MI) Scores

Collocation analysis identifies words that co-occur with a target keyword more frequently than chance (Sinclair 1991). MI score > 3.0 is considered a significant collocation. The collocational profiles below are compared across periods to detect meaning drift through changing lexical environments.

Target Keyword	Collocate	Window	Co-freq	MI Score	Period	Drift Signal
community	solidarity	L5–R5	[N]	[4.2]	P1	Anchor collocate
community	protect	L5–R5	[N]	[3.8]	P2	NEW — EXCLUSION shift
community	values	L5–R5	[N]	[5.1]	P2	NEW — boundary marker
power	shared	L5–R5	[N]	[4.6]	P1	Anchor collocate
power	struggle	L5–R5	[N]	[3.9]	P2	NEW — CONFLICT shift
[keyword]	[collocate]	[win]	[N]	[X.X]	[P]	[signal]

4.3 Type-Token Ratio (TTR) Across Corpus Sections

TTR (unique word types / total tokens) measures lexical diversity. A declining TTR in Period 2 indicates increasing lexical repetition — a marker of rhetorical consolidation or formulaic register (Sinclair 1991).

Corpus Section	Tokens (N)	Types (N)	TTR	Interpretation
Period 1 (full)	[N]	[N]	[0.XX]	Baseline lexical diversity
Period 2 (full)	[N]	[N]	[0.XX]	Comparative TTR — drift indicator
Period 1 (target words only)	[N]	[N]	[0.XX]	Keyword-focused lexical range
Period 2 (target words only)	[N]	[N]	[0.XX]	Keyword-focused lexical range
Overall Corpus	[N]	[N]	[0.XX]	Full corpus TTR for reference

SECTION 5 — CROSS-AGENT SYNTHESIS & THEORETICAL INTERPRETATION (Agent 5)

Role of Agent 5: The Orchestrator-Reporter does not generate new analysis. It evaluates convergence and divergence across Agents 1–4, applies Critical Discourse Analysis (Fairclough 1992) as the overarching interpretive lens, and constructs theory-grounded explanations for each finding. Where agents produce sparse or contradictory outputs, Agent 5 initiates a targeted re-analysis loop.

5.1 Evidence Convergence Matrix

The convergence matrix tracks which findings are corroborated by multiple agents. A finding supported by 3 or more independent agents is classified as **High-Confidence Evidence** for the research record.

Finding	A2 Pragmatic	A3 Semantic	A4 Statistical	Confidence	Theoretical Basis
'Community' → EXCLUSION field shift	✓	✓	✓	HIGH	Lyons (1977); Fairclough (1992)
Directive speech acts increase P2	✓	—	✓	MEDIUM	Searle (1969); Grice (1975)
Maxim Quantity violations +22% P2	✓	—	✓	MEDIUM	Grice (1975, Maxims)
Register borrowing: formal→informal	—	✓	✓	MEDIUM	Halliday (1978, Register)
Politeness decline (face-threat ↑)	✓	✓	—	MEDIUM	Brown & Levinson (1987)
[Additional finding]	[A2]	[A3]	[A4]	[Level]	[Citation]

Table 5.1: Cross-agent convergence matrix. ✓ = corroborating evidence present; — = not applicable to agent's analytical domain. HIGH confidence = 3 agents converge; MEDIUM = 2 agents; LOW = 1 agent only.

5.2 Theoretical Framework Alignment

Agent 5 maps each major finding to its supporting theoretical literature. This section constitutes the bridge between corpus-derived evidence and theoretical claims — the core requirement for academic publication.

Theoretical Framework	Key Concept Applied	Finding Supported	Corpus Evidence Count
Grice (1975) Cooperative Principle	Maxim flouting generates implicature	Pragmatic drift via strategic violation	[N] segments
Austin (1962) / Searle (1969) Speech Acts	Illocutionary force classification	Directive increase in Period 2	[N] segments
Brown & Levinson (1987) Politeness	Face-threatening acts; negative pol.	Declining politeness scores P2	[N] segments
Lyons (1977) Semantic Fields	Lexical field membership & migration	Semantic drift for target keywords	[N] word types
Halliday (1978) SFL Register	Tenor / Field / Mode analysis	Register borrowing detected	[N] segments

Theoretical Framework	Key Concept Applied	Finding Supported	Corpus Evidence Count
Sinclair (1991) Corpus Linguistics	Collocation, frequency, keyness	Statistical confirmation of all findings	[N] collocations
Fairclough (1992) CDA	Power/ideology in discourse	Ideological function of drift patterns	Synthesis of all

SECTION 6 — EVIDENCE RATIO FRAMEWORK & METHODOLOGY

This section defines the mandatory evidence ratio framework applied throughout this report. The framework ensures that the PMDD output meets the evidentiary standards required by the linguistic research community, distinguishing it from interpretive AI narrative generation.

6.1 Quantitative vs Qualitative Evidence Ratio

Report Section	Quantitative Evidence (Statistical, Measurable)	Qualitative Evidence (Interpretive, Corpus-cited)	Min. Corpus Examples Required per Claim
Executive Summary	40%	60%	2 per major finding
Pragmatic Analysis (Sec. 2)	55%	45%	3 per speech act category
Semantic Field Analysis (Sec. 3)	40%	60%	3 per drifting keyword
Corpus Statistics (Sec. 4)	85%	15%	Tables serve as evidence
Cross-Agent Synthesis (Sec. 5)	30%	70%	1 per theoretical claim
OVERALL REPORT TARGET	50%	50%	Min. 3 per major finding

Table 6.1: Mandatory evidence ratio targets per report section. These ratios are enforced by Agent 5's quality gate before report generation. Sections below threshold trigger automatic re-analysis of the responsible agent.

6.2 Corpus Evidence Presentation Standard (3-Layer Protocol)

Layer	Content	Format in Report	Purpose
Layer 1: Raw Corpus	Original segment text from corpus, with segmentation tags	Segmented block with SEG # and SEG type	Primary evidence — verifiable by any reader
Layer 2: Agent Annotation	Agent's classification, scores, and tags	Applied to segmented quote in evidence block	Links raw text to analytical framework
Layer 3: Theory Interpretation	Agent's explanation using named theoretical framework	Attached paragraph citing author & year	Grounds finding in peer-reviewed linguistics

Table 6.2: The 3-Layer Corpus Evidence Protocol applied to all qualitative findings in Sections 2, 3, and 5. This protocol guarantees traceability from interpretation back to primary corpus data — essential for peer review reproducibility.

6.3 Adaptive Agent Methodology — Not Rule-Based

Unlike traditional NLP pipelines, PMDD agents do not apply fixed linguistic rules. Each agent uses GPT-4o as its reasoning backbone with carefully designed system prompts that specify theoretical frameworks as interpretive lenses rather than classification dictionaries. This adaptive approach enables agents to:

- **Identify indirect and embedded speech acts** — not pre-defined in Searle's taxonomy
- **Detect emergent semantic fields** — that arise from the corpus itself, not pre-mapped ontologies
- **Recognise novel register borrowing patterns** — including cross-genre and cross-period hybrids
- **Update collocational profiles** — based on episodic memory from previous corpus analyses
- **Flag analytically ambiguous cases** — for human researcher review rather than forcing classification

Episodic Memory Note: Agents store previous analysis sessions in a persistent vector database. On each new corpus run, Agent 5 retrieves the three most similar prior analyses and includes pattern comparisons in the synthesis. This enables longitudinal research tracking across multiple corpus uploads by the same research team. Memory entries include: corpus metadata, top findings, drift scores, and theoretical framework usage frequency — enabling adaptive theory selection for future analyses.

SECTION 7 — RESEARCH IMPLICATIONS & CONCLUSIONS

The PMDD system produces findings that are directly usable in academic research contexts. The conclusions below are structured to address the key research questions posed at corpus submission, with each claim supported by evidence references to earlier sections of this report.

7.1 Primary Research Findings

Finding No.	Claim	Evidence Source	Confidence	Theoretical Basis
F-01	Pragmatic meaning drift is confirmed at [XX]/100 — above the 60-point research threshold	Table 2.1; Sec. 2.2; Seg. #203, #445, #612	HIGH	Grice (1975); Searle (1969)
F-02	Target keyword 'community' has undergone semantic field migration from SOLIDARITY to EXCLUSION	Table 3.1; Sec. 3.1; Table 4.2	HIGH	Lyons (1977); Fairclough (1992)
F-03	Directive speech acts show statistically significant increase in Period 2 (p < 0.05)	Table 2.1; Keynes Table 4.1	MEDIUM	Searle (1969)
F-04	Register borrowing: formal political vocabulary detected in informal contextual segments	Table 3.2; TTR Table 4.3	MEDIUM	Halliday (1978)
F-[N]	[Additional finding from this corpus]	[Evidence ref]	[Level]	[Theory]

7.2 Limitations & Future Research

Limitation	Methodological Note	Suggested Future Work
Corpus size	Results are corpus-specific; generalisation requires replication	Quota sampling of comparable corpora
GPT-4o variability	LLM temperature=0.2 minimises variance; no dissimulated	Cross-model validation with Gemini/Claude
Temporal granularity	Two-period split is a methodological simplification	Sub-period analysis for finer drift tracking
Language scope	Current system optimised for English L1 corpora	Multilingual agent extension required

APPENDIX A — AGENT ARCHITECTURE & EPISODIC MEMORY DESIGN

Agent	LLM Backbone	Memory Type	Vercel Deployment	Persistent Storage	Re-analysis Trigger
Agent 1: Preprocessor	GPT-4o (temp 0.0)	In-context only	Serverless Function	Corpus metadata JSON	File format error
Agent 2: Pragmatic	GPT-4o (temp 0.2)	Episodic (top-3 similar)	Serverless Function	Annotation vectors DB	<50% segment coverage
Agent 3: Semantic	GPT-4o (temp 0.3)	Semantic embeddings	Serverless Function	Semantic field clusters	Drift map < 5 words

Agent	LLM Backbone	Memory Type	Vercel Deployment	Persistent Storage	Re-analysis Trigger
Agent 4: Statistician	Python (no LLM)	None (stateless)	Serverless Function	Frequency tables CSV	Statistical threshold fail
Agent 5: Orchestrator	GPT-4o (temp 0.1)	Episodic + Semantic	Serverless Function	Full report + scores	Any agent below QC gate

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